

Industrial process chillers

WCP series

WCP chillers keep process water at a prescribed set temperature using refrigeration. Each unit includes a tank, circulating pump and air- or water-cooled condenser. Tank material depends on the model and order specification; see the model table.



Capacity	Temperature difference	Inlet water pressure	Inlet water temperature
0.5 to 20 TR	5 °C	2 to 4 bar g	5 to 30 °C

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Specifications online

Layout edition 2026-10 · Product-range information. Confirm the selected model and duty with Win Equipments before ordering.

Models & rating conditions

Rating conditions

Water outlet temperature (ideal)	15 °C	Ambient temperature (ideal)	40 °C
Water ΔT	5 °C	Water inlet pressure	1 bar g ideal · 2–4 bar g max

Capacity & operating data

Model	Capacity TR	Capacity m³/hr	Refrigeration compressor kW	Electrical V / Ph / Hz	Cooling fans qty	Condenser
WCP 005	0.5	0.3	0.9	230 / 1 / 50	1	Air / Water
WCP 007	0.75	0.45	1.2	230 / 1 / 50	1	Air / Water
WCP 010	1	0.6	1.5	230 / 1 / 50	1	Air / Water
WCP 020	2	1.2	2.5	230 / 1 / 50	1	Air / Water
WCP 030	3	1.8	3.4	440 / 3 / 50	1	Air / Water
WCP 050	5	3.0	6.0	440 / 3 / 50	2	Air / Water
WCP 075	7.5	4.5	9.2	440 / 3 / 50	3	Air / Water
WCP 100	10	6.0	11.5	440 / 3 / 50	4	Air / Water
WCP 150	15	9.0	18.0	440 / 3 / 50	6	Air / Water
WCP 200	20	12.0	22.5	440 / 3 / 50	8	Air / Water

Construction, dimensions & connections

Model	Water flow LPM	Water pump HP	Water tank litres	Tank material	Shipping weight kg	End connection BSP / NB
WCP 005	6	0.5	15	SS	125	¾"
WCP 007	8	0.5	20	SS	135	¾"
WCP 010	10	0.5	25	SS	150	¾"
WCP 020	21	0.75	40	SS	200	1"
WCP 030	32	1	60	SS	240	1"
WCP 050	52	1.5	80	SS	350	1½"
WCP 075	80	2	140	SS	475	1½"
WCP 100	100	2	180	SS / MS	550	2"
WCP 150	155	2	250	SS / MS	700	2"
WCP 200	202	2	320	SS / MS	1000	2"

- Model code: WCP + capacity in TR (e.g. WCP 050 = 5 TR). Working pressure 2–4 bar g max.
- Pressure drop within 0.2 bar g max.
- Rated power is the maximum power consumed at rated conditions.
- Voltage range 180–260 V for 1 Ph and 380–420 V for 3 Ph.

How it works & construction

How it works

The chilling plant has two circuits: a refrigeration system (compressor, condenser, fan, expansion device and evaporator) and a water circulating system (pump, motor and tubing).

1 Water in

Process water returns to the chiller through the water inlet.

2 Evaporate

In the brazed plate heat exchanger (evaporator) the water is cooled by the refrigerant. Counter-flow achieves the maximum temperature difference with minimum pressure drop.

3 Reject heat

The refrigerant compressor sends the absorbed heat to an air- or water-cooled condenser.

4 Water out

Cooled water is pumped from the built-in tank back to the process at the set temperature.

Construction & features

Brazed plate heat exchanger

- Sized to keep fluid velocities and turbulence high
- Counter-flow for maximum temperature difference
- Minimum pressure drop

Circulating pump

- Stainless steel impeller, mechanical seal
- Rated for 24-hour duty
- Motor overload protection

Controls

- Programmed temperature controller with alarms
- On-delay protects the compressor
- High / low pressure gauges and switches

Refrigeration compressor

- Hermetically sealed reciprocating or scroll
- Suitable for eco-friendly gases
- Low noise, low power consumption

Tank

- Tank material as listed for the model: SS or SS / MS
- Closed tank: no heat gain
- Removable steel covers for access

Hot gas bypass valve

- Prevents evaporator temperature dropping below 5 °C
- Manual or automatic

Selection & applications

Sizing for your conditions

For water: heat load (kcal/hr) $\approx 60 \times$ water flow (L/min) $\times$ (inlet – outlet temperature, °C). This is a water approximation; confirm the cooling duty and rating conditions before selection.

Water outlet temperature

°C	Factor
5	0.6
10	0.75
15	1.0
20	1.16
25	1.3

Ambient temperature

°C	Factor
30	1.2
35	1.1
40	1.0
45	0.9
50	0.8

Applications: Automobile · Injection moulding & plastics · Hydraulic presses & drives · Compressor cooling · Milk, dairy & pasteurising · Food products · Lubricant & oil cooling · Pulp & paper · Printing ink & paper waxing · Sugar · Footwear

What two values do I need to select a chiller?

The required water outlet (cold) temperature, and the heat load to be removed. Heat load $Q =$ mass flow (LPM) $\times \Delta T$ between inlet and outlet temperature.

What is the tank made of?

Up to WCP 075 the tank is stainless steel. WCP 100 to WCP 200 are offered in stainless or mild steel.

Are WCP chillers air cooled or water cooled?

Every WCP model is available with either an air-cooled or a water-cooled condenser.

How does ambient temperature affect capacity?

Capacity is rated at 40 °C ambient. At 45 °C apply a factor of 0.9, at 50 °C 0.8.

Confirm your selection

Send the model, required capacity, operating temperatures, pressure and site conditions. For a model-specific datasheet or GA drawing, include your installation requirement.

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Catalogue values are for product selection. Confirm duty, materials, connections and final dimensions with Win Equipments before ordering. Layout edition 2026-10.